A COUNTABLE RING IS COHERENT IFF R^N IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The proof leans on the two preceding examples in the same section (fg-tensor and fp-tensor lemmas); both are delivered cards in this batch, so the cluster is self-contained.

Editorial note. The label lemma-countable-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Coherence: every finitely generated ideal is finitely presented; Algebra, proposition-characterize-coherent (coherent rings characterized by flatness of R^N / products)
- Same-chapter cluster: lemma-countable-fg-tensor (surjectivity detects finite generation) and lemma-countable-fp-tensor (isomorphy detects finite presentation)

Source reference. Original proposition 05CZ: sources/stacks-chow/tags.txt, source lines 6937–6937. Prerequisite pointer invoked at 1008; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AL9: sources/stacks-examples/pinned-original-source.tex, source lines 963–963. Prerequisite pointer invoked at 1014; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ALA: sources/stacks-examples/pinned-original-source.tex, source lines 980–980. Prerequisite pointer invoked at 1015; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a countable ring. Then R is coherent if and only if $R^{\mathbf{N}}$ is a flat R-module.*

Proof. If R is coherent, then $R^{\mathbf{N}}$ is a flat module by Algebra, Proposition 05CZ. Assume $R^{\mathbf{N}}$ is flat. Let $I \subset R$ be a finitely generated ideal. To prove the lemma we show that I is finitely presented as an R-module. Namely, the map $I \otimes_R R^{\mathbf{N}} \to R^{\mathbf{N}}$ is injective as $R^{\mathbf{N}}$ is flat and its image is $I^{\mathbf{N}}$ by Lemma 0AL9. Thus we conclude by Lemma 0ALA. □

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.